Optimizing Online Retail Sales performance using a Data-Driven Approach

A Final report for the BDM capstone Project

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1. Executive Summary

The Final report of this capstone project shows the detailed analysis, results and insights into the problems of an e-commerce retailer based in the United Kingdom, known for its curated selection of home decor and gift items serves to both individual shoppers(B2C) and wholesale buyers(B2B) worldwide. In recent years, the company has witnessed fluctuating demand, seasonal sales variations, and increasing return rates**,** which have created significant challenges in inventory management and operational planning. While the business has achieved widespread reach, the company encounters challenges, it continues to grapple with operational inefficiencies that impact its growth and profitability.

The pain points include inconsistent demand, unclear insights into product-level performance, and a high rate of product returns. Unpredictable sales cycles often lead to excess inventory or stockouts, and a significant share of sales is driven by only a handful of items in the catalog. Additionally, frequent returns not only reduce profitability but also signal potential issues in product quality or customer expectations, highlighting the need to analyze return behaviors more thoroughly.

To address these issues, the proposed solution is grounded in data-analytics. The project will employ time-series forecasting models to improve demand accuracy and help maintain optimal inventory levels. A Pareto-based product analysis will be conducted to spotlight top-performing items, informing both marketing strategies and stocking decisions. Moreover, an in-depth review of return patterns will offer actionable insights into customer dissatisfaction and guide quality control efforts.

2. Proof of Originality

The dataset used for this project titled Online Retails Sales Data was sourced from an open access repository UCI Machine Learning Repository a reputable platform for data sharing. The data includes crucial variables such as order details, customer details, etc. The dataset was downloaded directly from the source and has not been modified or manipulated externally, ensuring the originality and integrity of the data for analysis.

The link for the analysis notebook- Analysis
The link for the dataset - Dataset

3. Meta data and descriptive Statistics

- The dataset contains 541,909 transaction records from a UK-based online retailer, covering the period December 1, 2010, to December 9, 2011**.**
- The dataset provides detailed records of order details**,**products, quantities, pricing, customers, and countries of purchase, enabling in-depth analysis of sales trends, customer behavior, and return patterns.

- The columns in the dataset are as follows :

Column Name	**Datatype**	**Description**
InvoiceNo	Categorical	a 6-digit integral number uniquely assigned to each transaction. If this code starts with letter 'c', it indicates a cancellation
StockCode	Categorical	a 5-digit integral number uniquely assigned to each distinct product
Description	Categorical	product name
Quantity	Integer	the quantities of each product (item) per transaction
InvoiceDate	Date	the day and time when each transaction was generated
UnitPrice	float	product price per unit
CustomerID	float	a 5-digit integral number uniquely assigned to each customer
Country	Categorical	the name of the country where each customer resides

- Below is a summary statistics of the dataset:

	Quantity	**InvoiceDate**	**UnitPrice**	**CustomerID**
count	541909.00000	541909	541909.000	406829.00000
mean	9.552250	2011-07-04 13:34:57	4.611114	15287.00000
std	-80995.000	2010-12-01 08:26:00	-11062.060	12346.00000
min	1.00000	2011-03-28 11:34:00	1.250000	13953.00000
25%	3.00000	2011-07-19 17:17:00	2.080000	15192.00000
50%	10.00000	2011-10-19 11:27:00	4.130000	16791.00000
75%	80995.00000	2011-12-09 12:50:00	38970.0000	18287.00000
max	218.081158	NaN	96.759853	1713.600303

- Data Quality Overview:
  i. The dataset contains 1,454 missing values in Description column and 135,080 missing values in CustomerID column.
  ii. A total of 5,268 duplicate records were identified.

4. Detailed explanation of analysis process/method

1. **Data Preparation ,Cleaning and Data Extraction.**

   The dataset required several preprocessing steps to prepare it for analysis :
   - The Description and CustomerID columns contained missing values which were addressed as follows :
     - Missing descriptions were replaced with 'Unknown'.
     - Missing customer IDs were replaced with 'Unknown_customer'.
   - There were 5268 duplicated rows which were removed to eliminate redundancy and maintain data integrity.
   - Additionally, new features were engineered into the dataset to enable deeper analysis and enhanced insights :
     - Revenue Calculation : Introduced a new column into the dataset by multiplying Quantity and UnitPrice to represent the transaction value.
     - Date-Based Features : Extracted additional time-based features such as Year,Month,Day and Weekday from the InovoiceDate column to support trend analysis.

2. **Pareto analysis**

   The goal of this analysis was to determine which products and customers were most influential in generating total revenue . This approach is guided by the Pareto principle (80/20 rule), which suggests that a small proportion of items or customers often contribute disproportionately to overall business results.
   - **Data Organization**
     - Compiled total sales revenue by product and by customer.
     - Sorted these lists in descending order of revenue to prioritize top contributors.
   - **Applying the Pareto Concept**
     - Calculated the cumulative share of total revenue across products and customers.
     - Identified which top-ranked items and customers accounted for approximately 80% of total revenue.
   - **Key Findings**
     - A small group of products was responsible for the bulk of total sales, indicating that focusing on these products can drive significant revenue.
     - Similarly, a small subset of customers contributed to the majority of purchases, highlighting the importance of tailored marketing and loyalty strategies for this group.

- **Visualization of Results**
  - Created Pareto charts showing the cumulative revenue share contributed by products and customers, with clear indicators of where the 80% threshold was reached.
  - Separate visualizations were made for product and customer revenue contributions to better illustrate the distribution.

Summary :

i. Approximately 20% of products generated around 80% of total sales revenue. Prioritizing these products for inventory and sales focus could optimize business resources.

ii. About 20% of customers were responsible for most purchases, suggesting that focusing marketing efforts and loyalty programs on this customer segment can yield substantial returns.

3. Demand Forecasting

The goal of demand forecasting is to utilize past sales data to predict upcoming sales trends. By doing so, it is aimed to improve inventory management by ensuring optimal stock levels and reducing the risk of overstocking or stock shortages.

- **Data Preparation**
  - Extracted sales data over a defined period.
  - Aggregated total sales figures at both daily and monthly intervals to facilitate analysis.

- **Exploratory Data Analysis (EDA) – Identifying Patterns**
  A series of exploratory analysis were carried out to identify the underlying hidden patterns and trends within the sales data :
  - **Visual exploration** : Line plots and time series graphs were used to visualize how sales data evolved over time. These plots helped in understanding data structure.
  - **Autocorrelation analysis** :
    - **Autocorrelation Function (ACF)** :
      This technique measured the correlation between current sales values and past observations . For example, if the data contained seasonal effects (e.g., higher sales every December), the ACF plot would show recurring peaks at intervals matching the seasonality period.
    - **Partial Autocorelation Function** :
      While ACF measured total correlations ,PACF focused on direct relationships between current and lagged values. Sharp drops in PACF after specific lags suggested that how many times points influenced the current value
  - **Testing for Stationarity** :
    Time series forecasting models assume that the data is stationary, its statistical properties like mean and variance does not change over time.

To validate this, the **Augmented Dickey-Fuller (ADF) test** was applied. This checked whether the sales data exhibited unit roots, which are indicators of non-stationarity.
The test results revealed that the original data was **non-stationary**, characterized by a gradual upward trend and fluctuating variance

- **Data transformation to achieve Stationarity :**
  To correct the non-stationarity and make the data ready for modeling, the **first-order differencing(d=1)** was applied. This involved calculating the difference between a given month's sales and the sales of the previous month. The differencing removed the trend and maintained the variance, making the dataset stationary and ready for ARIMA modeling.

- **Model Development and Parameter Selection**
  Using insights from the earlier exploratory analysis and stationarity checks:
  - **Autoregressive component (p)**: Set to 0, as PACF plots indicated no significant lag-based correlations in the differenced data.
  - **Differencing order (d)**: Set to 1, reflecting the single level of differencing applied to make the data stationary.
  - **Moving average component (q)**: Set to 0, since the ACF plots did not reveal notable moving average behavior. Although these settings suggested a simple ARIMA(0,1,0) model without complex autoregressive or moving average terms, the model was implemented to assess its forecasting effectiveness.

- **ARIMA Model Training, Testing and Evaluation**
  To evaluate the model's performance, the data was divided into **training** and **test** sets. The ARIMA model was trained on the historical data. Three performance metrics were used:
  - Mean Absolute Error (MAE)
  - Mean Squared Error (MSE)
  - Root Mean Squared Error (RMSE)

4. Return analysis

Product returns can significantly impact overall revenue, so the focus of this analysis was to pinpoint the products and customers responsible for frequent returns and substantial revenue losses. The analysis also aimed to identify whether return volumes exhibit seasonal fluctuations. Gaining these insights can support better management of returns and help reduce revenue loss.

- **Data Extraction and Preparation :**
  - Filtered the dataset to isolate transactions marked as returns.
  - Created a new column, 'Revenue_Lost', to quantify the impact of each return.
- **In-Depth Analysis:**
  - **Identifying Products with High Return Rates**: Counted the number of returns for each product and ranked them to highlight the Top 10 returned items.

- **Highlighting Products with Significant Revenue Loss**: Summed up the total revenue loss for each product based on returns and plotted pie chart to identify the most financially impactful items.
- **Identifying Customers with High Return Activity**: Counted the frequency of returns per customer and sorted the data to reveal the most frequent returners.
- **Country-Level Analysis**: Created a bar chart to showcase the top 10 countries incurring the highest revenue losses due to product returns.
- **Examining Monthly Return Trends**: Plotted a line chart to visualize how revenue losses from returns evolved over time, revealing any seasonal or temporal patterns.

5. Customer Segmentation

The objective of the customer segmentation analysis was to identify unique customer groups based on their purchasing behavior. This segmentation enables businesses to implement more targeted marketing strategies, offer personalized promotions, and improve overall customer engagement and satisfaction.

- **Data Collection and Feature Selection**

To effectively group customers, we focused on key behavioral metrics:

- **Total Revenue**: The cumulative amount each customer spent on purchases.
- **Total Transactions**: The number of times a customer completed a purchase.
- **Total Products Purchased**: The total quantity of products bought across all transactions.
- **Unique Products Purchased**: The no. of product types bought by each customer.
- **Return Frequency**: The number of times a customer returned products.

- **Data Preprocessing**

Before clustering, we undertook essential preprocessing steps:

- Removed outliers, particularly the 'Unknown_Customer', which was excluded due to data inconsistencies that previously affected clustering accuracy.
- Applied Standard Scaler normalization to scale all features within a uniform range, ensuring that no feature disproportionately influenced the clustering process.

- **Clustering Approach**

We utilized K-Means clustering to categorize customers based on spending behavior and purchase frequency. To determine the optimal number of clusters **(**k**),** we employed two techniques:

- **Silhouette Score**:
  - Assessed the clustering quality for each k value, with higher scores indicating better separation between clusters.
  - The highest silhouette score occurred at k = 2, suggesting the presence of two well-defined customer groups.

- **Davis-Boulden Index:**
  - Computed the DBI, where lower values represent better clustering with less overlap.
  - The lowest DBI also appeared at k = 2, reinforcing the silhouette analysis results.

Summary :

The clustering revealed two primary customer segments:

- **Regular Shoppers(k=0)**: Customers who make occasional purchases, have lower total spending, and exhibit minimal return behavior. These customers represent the core base of steady but low-volume shoppers.
- **High-Value Customers(k=1)**: This group is characterized by frequent purchases, high total spending, and a higher likelihood of returns. These customers are significant contributors to revenue but may also drive return-related challenges.

5. Results and Findings

5.1 Pareto Analysis

This analysis helped in identifying:

- Top 20% Products contributing to 80% revenue
- Top 20% Customers contributing to 80% revenue

1. Product Pareto Analysis

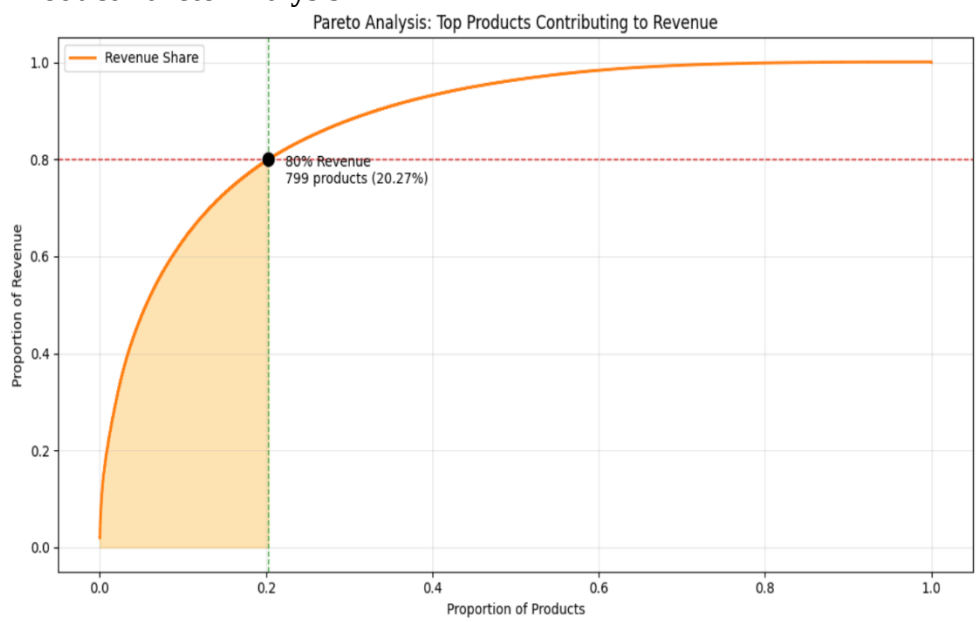

Fig 5.1.1 Product Pareto Curve

Insights:

a. Total Products :4070

b. Products that contributed to 80% revenue :799(20.27%)
c. Top 8 best-selling products :

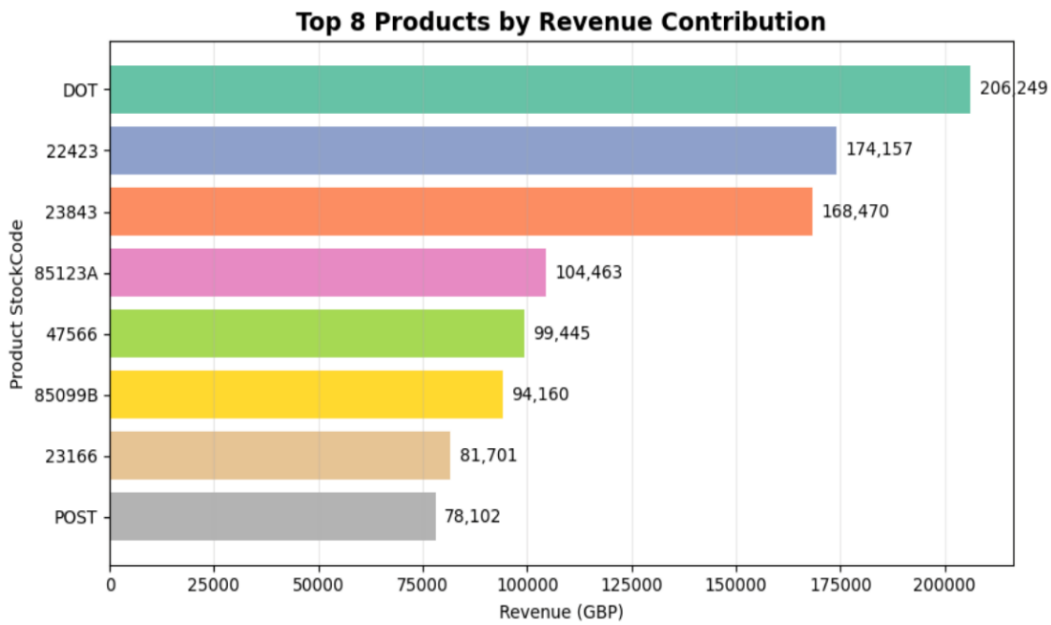

Fig 5.1.2 Top 8 revenue generating products

d. Since a limited number of products are responsible for most of the sales, so its important to focus inventory management.

2. Customer Pareto Analysis :

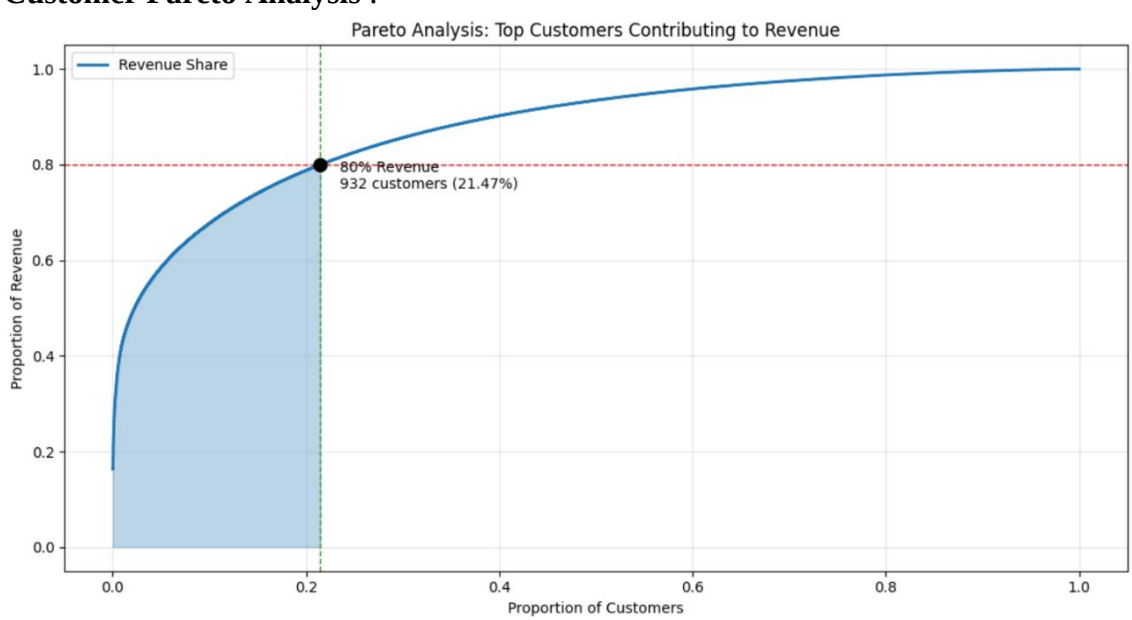

Fig 5.1.3 Customer Pareto Curve

Insights:

a. Total Customers : 4340
b. Total customers contributing to 80% revenue : 932(21.47%)
c. Here are the top 8 revenue generating customers(excluding Unknown_customer):

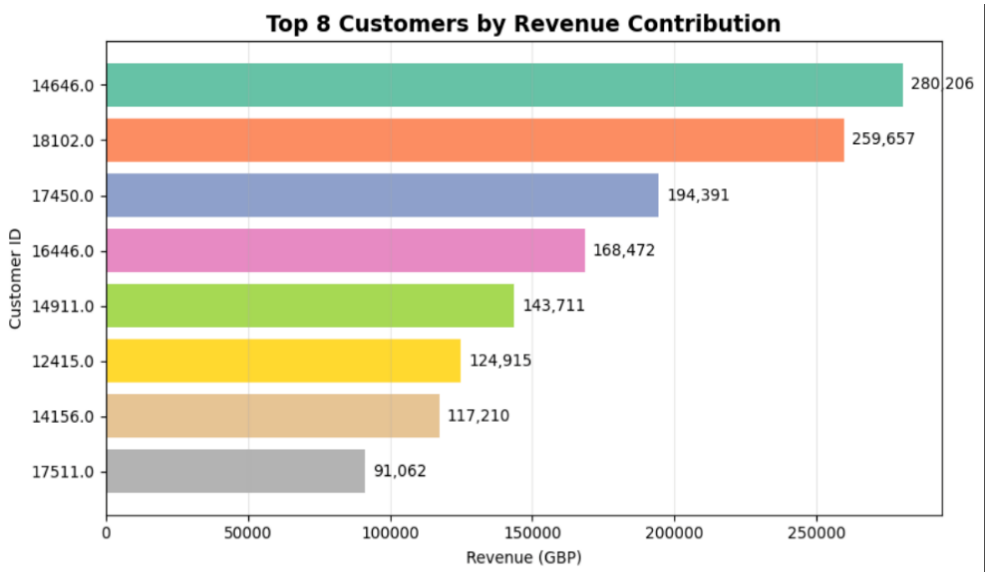

Fig 5.1.4 Top 8 revenue generating customers

d. A small segment of high-value customers contributes disproportionately to the overall revenue.

Results:

- A majority of sales is generated from a limited set of top-performing products. Focusing on inventory management can optimize stock levels and reduce inefficiencies.
- Identifying the top revenue generating customers enables more effective targeting in promotional campaigns,improving engagement and return on investment(ROI).

5.2 Demand Forecasting

1. Daily Revenue Performance Analysis :

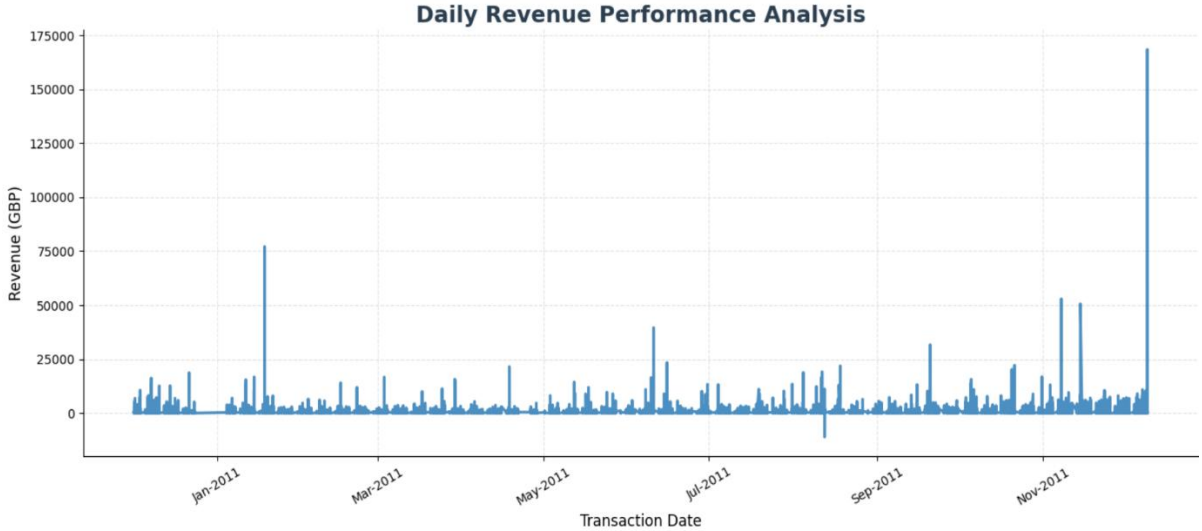

Fig 5.2.1 Daily sales trend analysis

Insights :

- **Significant Fluctuations** – Revenue patterns show frequent spikes, indicating inconsistent or unpredictable customer demand.
- **Presence of Outliers** – A few days exhibit unusually high sales, potentially driven by large business orders or limited-time promotional events.
- **Seasonal Trends and Lows** – Noticeable drops in sales occur on certain days, possibly due public holidays, or internal operational challenges.

2. Monthly revenue trend plot :

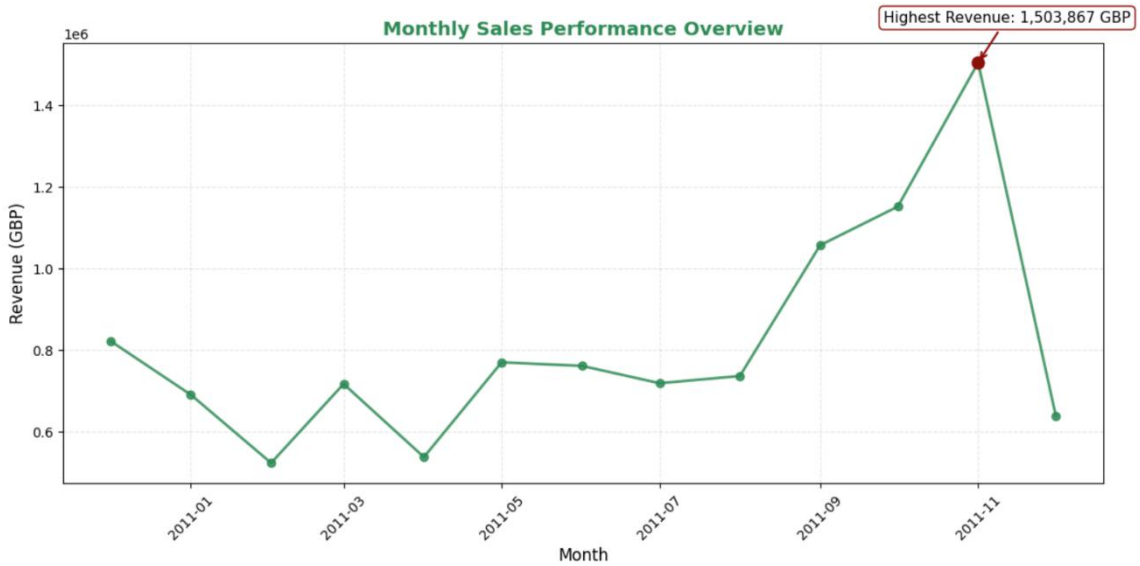

Fig 5.2.2 Monthly revenue trend analysis

Insights :

- **Cyclical Trends** – Revenue tends to dip at the beginning of the year and gradually increases, indicating a recurring seasonal pattern.
- **Year-End Boost** – A noticeable surge in sales occurs during November and December, likely driven by holiday-related shopping.
- **December Drop-Off** – A sudden decline toward the end of December may be due to post-holiday returns, decrease order or potential gaps in data collection.

3. **Checking for Stationarity :**

o ADF Statistic: -2.16
o P-value: 0.2196
o Critical Values:
- 1%: -4.137829
- 5%: -3.154972
- 10%: -2.71447

p-value (0.2196) is greater than 0.05, meaning that the time series is non-stationary. This suggests that it has trends or seasonality that we need to remove

4. **After Differencing the Data :**

o ADF Statistic: -3.68
o P-value : 0.004
o Critical Values:
- 1%: -4.22323
- 5%: -3.1893
- 10%: -2.7298

p-value is smaller than 0.05, meaning the time series is now stationary.

5. **Autocorrelation(ACF) vs Partial Autocorrelation(PACF) :**

- **Autocorrelation(ACF) plot :**

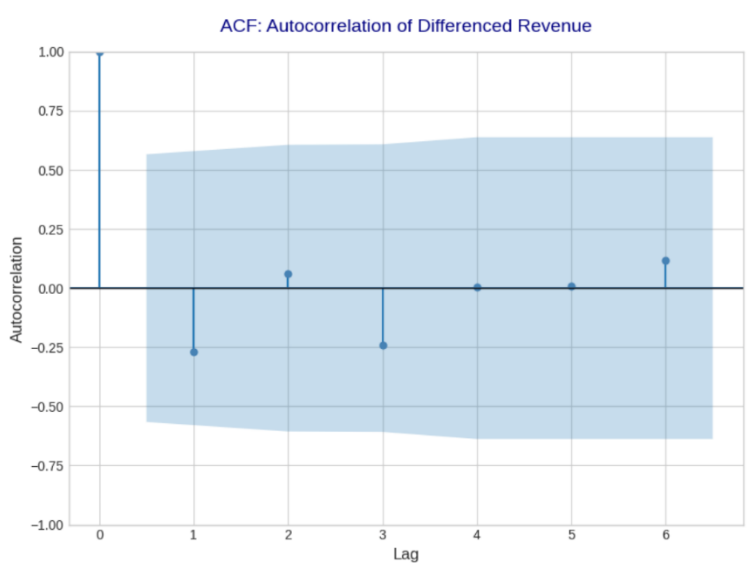

Fig 5.2.3 Autocorrelation(ACF) plot

- **Partial Autocorrelation(PACF) plot :**

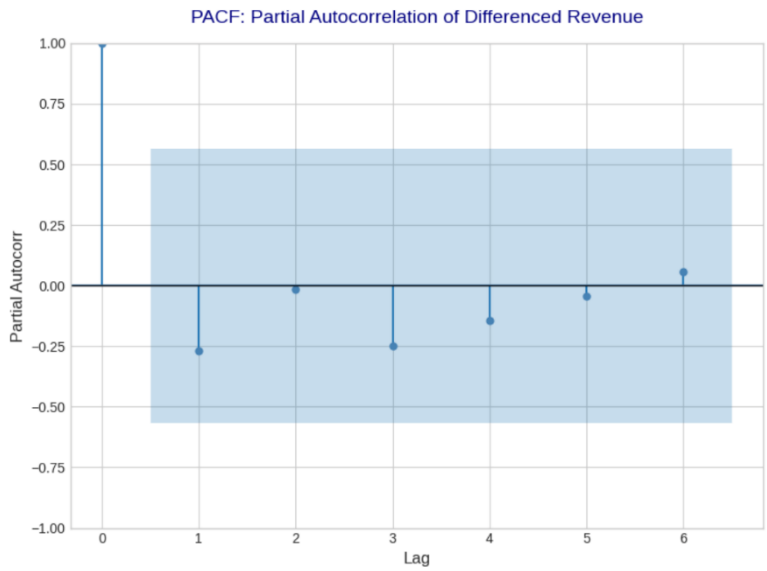

Fig 5.2.4 Autocorrelation(PACF) plot

Insights :

In both the ACF and PACF plots, all the lag values fall within the confidence interval (shaded region), indicating no statistically significant lags.

6. ARIMA Model :

- **Parameters for ARIMA:**
  - **'d' = 1 :** Applied first order differencing to make it stationary.
  - **'q' = 0** : ACF had no significant lags
  - **'p' = 0** : PACF had no significant lags.

- **Forecast vs Actual Plot :**

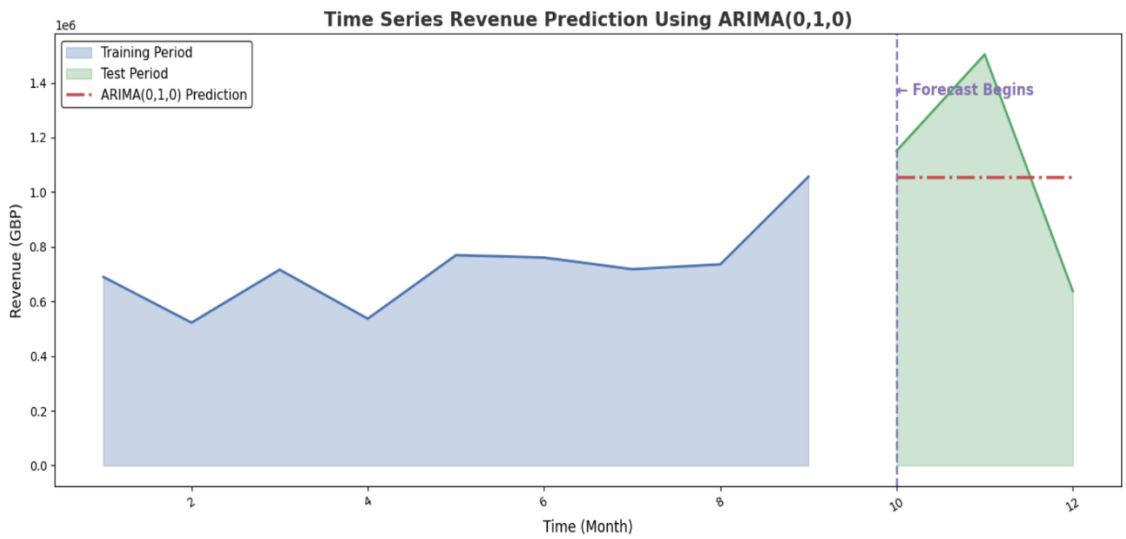

Fig 5.2.5 Forecast vs Actual Plot using ARIMA model

Insights:

- Limited data (12 months) restricted the model from capturing seasonal patterns.
- Forecasts showed high error and low predictive accuracy**.**
- Longer historical data is required for reliable ARIMA performance.

5.3 Return Analysis

Returned items accounts for approximately 9.2% of total sales revenue.

1. **Products with most returns:**

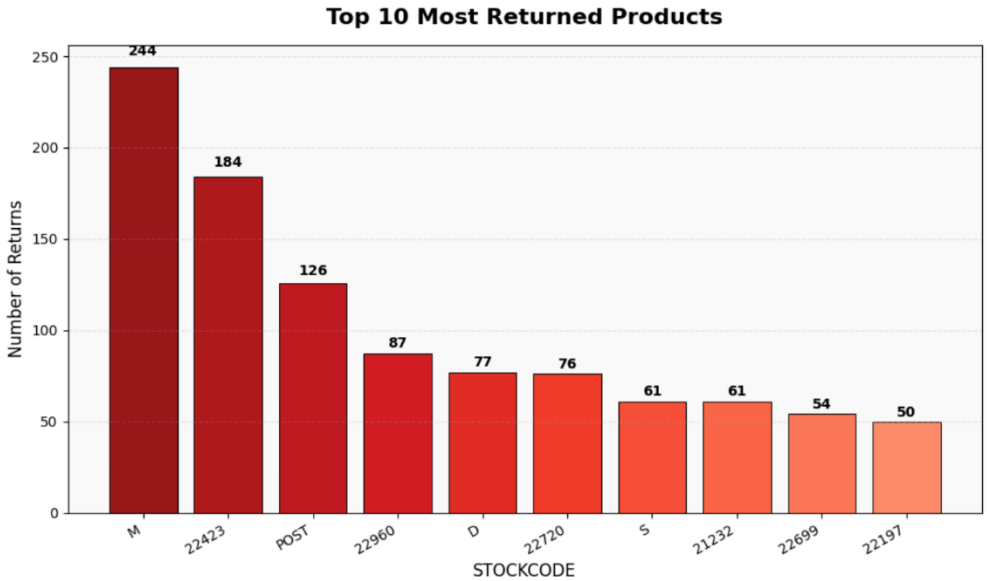

Fig 5.3.1 Top 10 products with most returns

- The product with stockcode ‘M’ is returned the most.

2. Top loss causing Products :

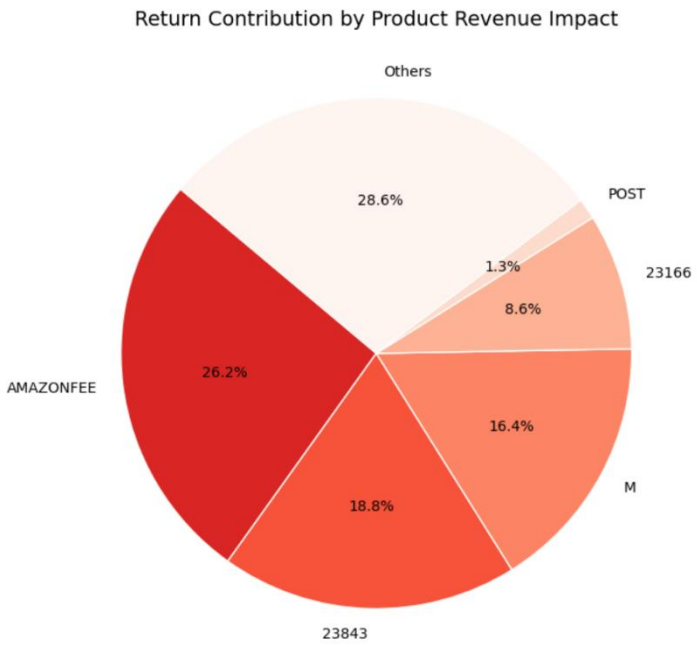

Fig 5.3.2 Top loss causing Products

- ‘AMAZONFEE has made the highest contribution in revenue loss.

3. Top Customers who returned the most:

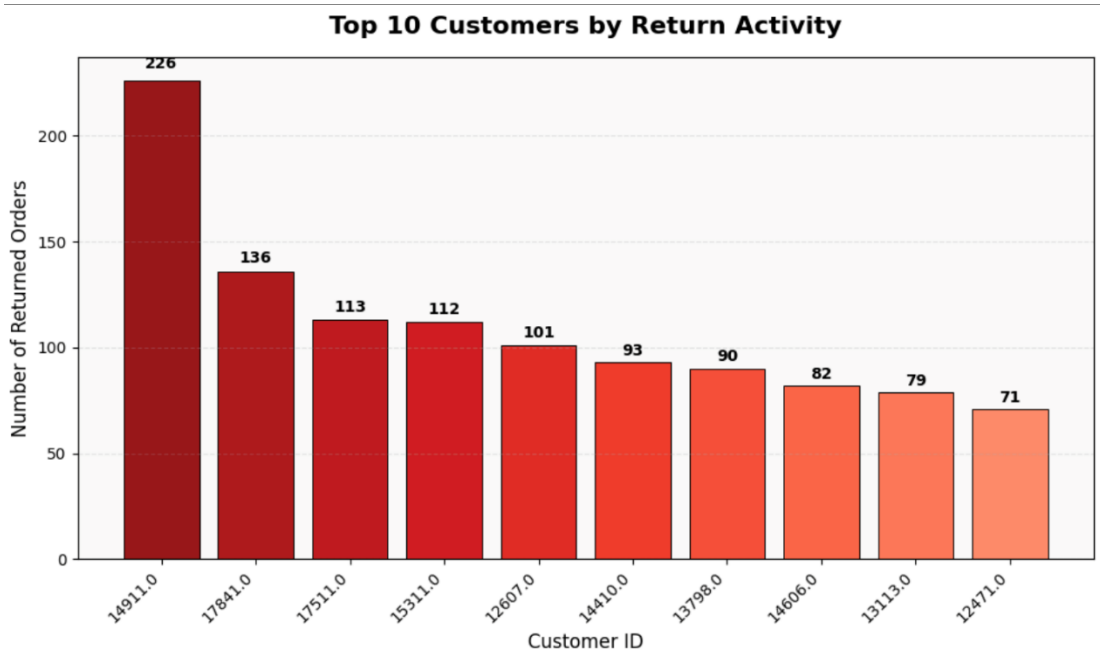

Fig 5.3.3 Top 10 customers who returned the most

- The customer with id ‘14911’ has made the most number of returns.

4. Country-wise analysis :

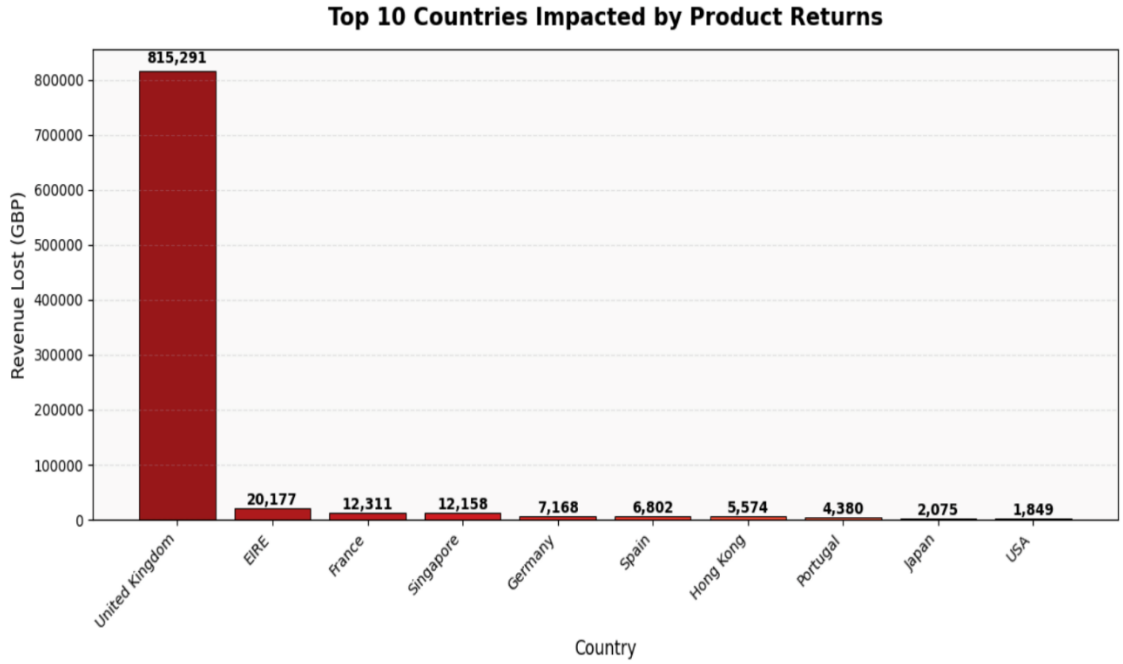

Fig 5.3.4 Top loss generating country

- Since UK is the primary market so it generated most loss and so it is expected.

5. Return patterns over the month :

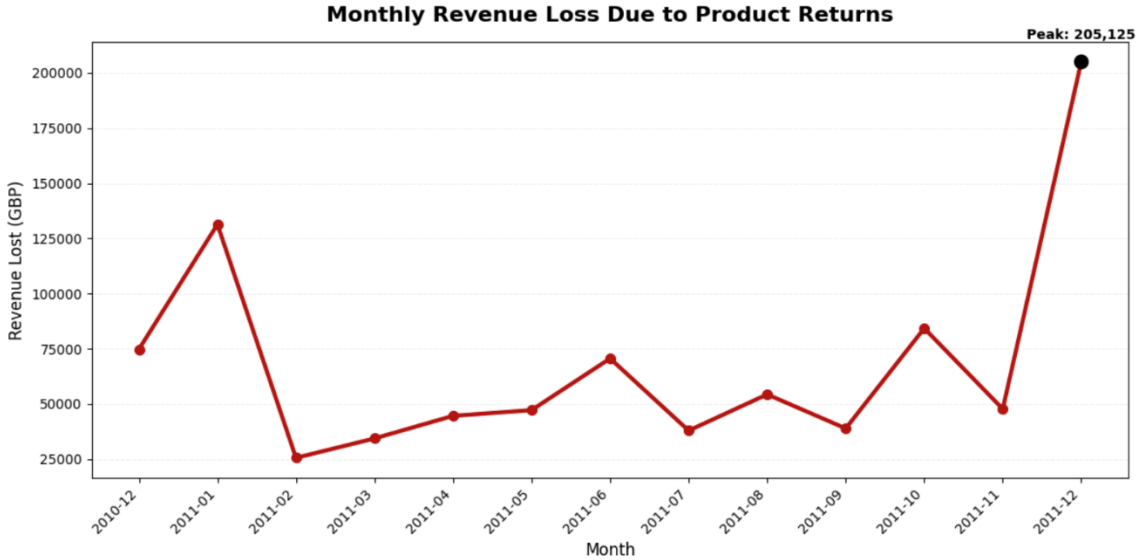

Fig 5.3.5 Return patterns over the month

- It is observed that most of the losses due to return is made in the month of November-January.

5.4 Customer Segmentation

After compiling customer-level data with key metrics - total revenue, Total transactions, Total products purchased, unique items purchased and return frequency and applying StandardScaler for normalization, the dataset was prepared and standardized for clustering analysis.

1. **Silhouette Method and Davies-Bouldin method to find the best number for k-means cluster.**

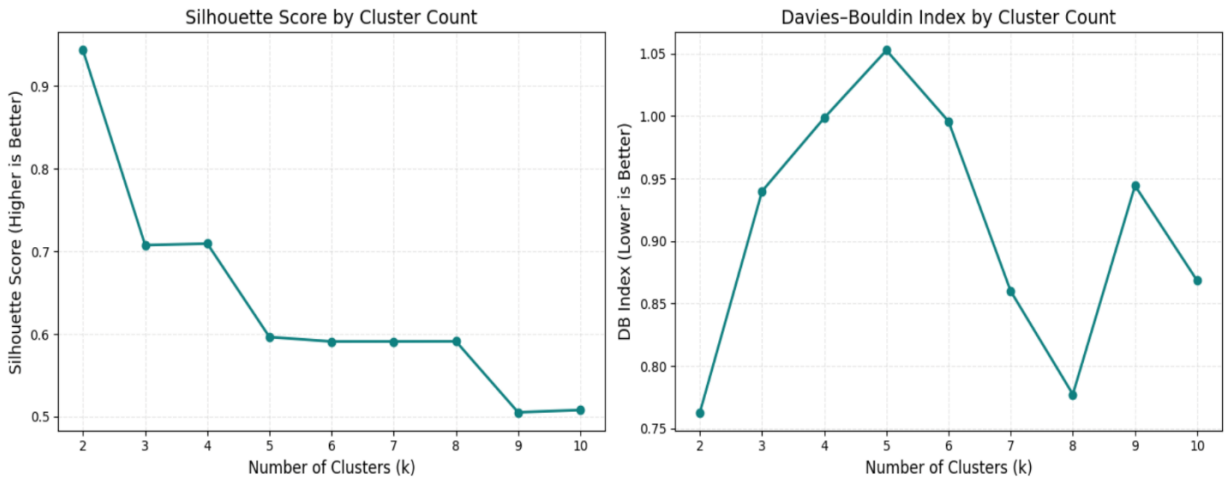

Fig 5.4.1 Silhouette Analysis and Davies-Boulden Analysis

Insights :

- Both of the methods suggested that k=2 is the best number of cluster for k-means.

2. **After applying k=2 clusters for k-means :**

Summary of clusters :

Cluster	Total Revenue	Total Transactions	Total Products Purchased	Unique Products Purchased	Return Frequency	Cluster	Cluster Size
0	1515.17	4.72	904.87	58.66	1.81	0.0	4354
1	94612.32	91.44	53727.78	677.56	56.57	1.0	18

3. **Clusters Visualization of customers segments:**

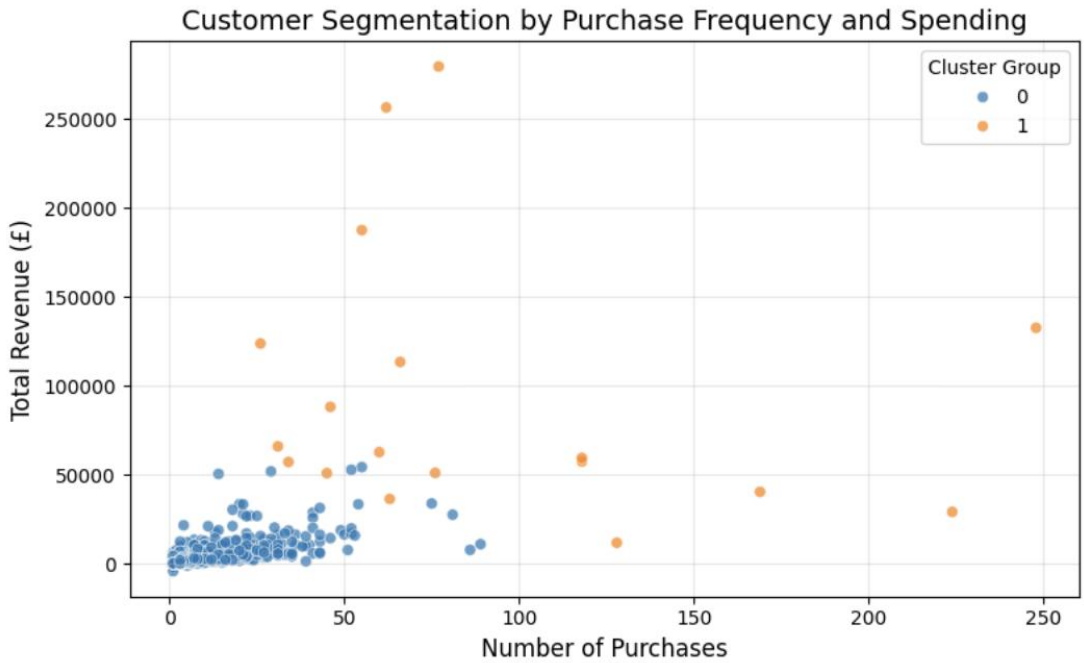

Fig 5.4.2 Cluster of customer segments

6. Interpretation of Results and Recommendation

6.1 Pareto Analysis

1. Top-Selling Products
   - High-revenue items included DOTCOM POSTAGE, Regency Cakestand 3 Tier, and Paper Craft + Little Birdie, alongside 820+ out of 4050+ products, contributing over 20% of total revenue.

   - **Recommendation**:
     Focus inventory planning, marketing and optimization on these top-performing products.

2. Top Customers
   - Key customers such as IDs 14646, 18102, and 17450, along with 930+ others (including unidentified users), generated over 20% of total revenue.

   - **Recommendations**:
     - Launch a loyalty program to retain high-value customers.
     - Resolve "Unknown_customer" issues by enhancing customer data collection.
     - Employ personalized marketing to boost engagement and sales.

6.2 Demand Forecasting

1. Daily Revenue Trend
   - Exhibited high variability with sharp spikes and dips, indicating inconsistent demand.
2. Monthly Revenue Trend
   - A cyclical pattern observed: revenue drops early in the year and peaks during November–December, likely due to holiday shopping.

Recommendations:
- Implement seasonal promotions to maximize peak-period revenue.
- Investigate underperforming months for possible operational inefficiencies.

ARIMA Model Performance

Model Metrics:
- MAE: 3,20,301.66
- MSE: 1,28,15,03,32,679.47
- RMSE: 3,57,980.91

Interpretation:
- Forecasting errors are too high, indicating that ARIMA may not be suitable for this dataset.
- The limited timeframe (12 months) lacks sufficient seasonality data for accurate forecasting.

Recommendations:
- Re-attempt forecasting with at least 2–3 years of data and external factors like holidays and promotions.
- Explore machine learning approaches such as Random Forest model or Lightbgm model for better predictive accuracy.

6.3 Return Analysis

Key Insights:
- Total Revenue Lost to Returns: GBP 896,812.49 (~9.2% of total sales).
- Peak Return Window: November to January, likely due to post-holiday returns.
- A small group of customers and UK-based sales contributed most to return losses.

Recommendations:
- Strengthen quality control for products with high return rates.
- Enforce stricter return policies during the holiday season.
- Offer special discounts on non-returnable items to high-return customers.
- Analyze return reasons to mitigate avoidable returns (e.g., unclear product descriptions).

6.4 Customer Segmentation

1. Cluster 0 – Regular Shoppers
   - Largest group; makes infrequent, low-spend purchases with low return rates

(avg. 1.81%).

Recommendations:
Use promotions and loyalty schemes to boost frequency and engagement.

2. Cluster 1 – High-Value Customers
   - High spenders with frequent and diverse purchases but a much higher return rate (avg. 56.57%).

Recommendations:
- Offer premium services such as early access, faster shipping, or dedicated support.
- Investigate reasons for high return rates and consider stricter return policies or enhanced product descriptions.

7. Presentation and legibility of the report

The project has been documented in a clear, organized, and visually appeal manner. Key aspects include:

- **Well-Structured Layout**: Each section of the report is systematically divided into sections—starting from data cleaning and exploration, through analysis and modeling, to interpretation and recommendations.
- **Consistent Formatting**: Headings, subheadings, bullet points, and tables are consistently formatted for easy navigation and quick understanding.
- **Readable Visuals**: Graphs, charts, and plots are clearly labeled and color-coded to enhance interpretability. All figures are accompanied by concise explanations to help the reader grab the insights at a glance.
- **Use of Clear Language**: Technical terms are used appropriately, and complex concepts are explained in simple language to maintain clarity for both technical and non-technical audiences.
- **Documentation**: The explanations use simple and precise language, making the workflow easy to follow and reproducible.

Overall, the project maintains standards of readability and presentation quality, ensuring that insights and conclusions are easy to read for the readers.